

- 9 The table shows three geostationary satellites, **P**, **Q** and **R**, and their masses.

satellite	mass of satellite / kg
P	1200
Q	1500
R	3500

Which satellite has the largest centripetal force?

- A P
 - B Q
 - C R
 - D They have the same magnitude of centripetal force
- C** geostationary satellites $\rightarrow$ same period T of 24 h $\rightarrow$ same r and ω
centripetal force $= mr\omega^2$
so mass is the factor. Biggest mass $\rightarrow$ largest centripetal force

- 10 A polystyrene cup contains a mass of 88 g of water at 20 °C. A cube of ice of mass